

Ngoc Quynh

Huong Tran

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PROFILE | CAREER OBJECTIVE

Creative and analytical software designer with expertise in Virtual Reality and Game Development, combining strong technical proficiency with immersive design, rapid prototyping, and problem-solving under pressure. Skilled in integrating physics-based systems, AI-driven mechanics, and user-centred interactions to create engaging and innovative digital experiences, with interests in LLM-based systems or agent workflows. Strong leadership, cross-cultural collaboration, and team coordination abilities, complemented by multilingual communication and a commitment to social impact. Passionate about green, sustainable technology and actively seeking solutions that are not just efficient but environmentally responsible. Known for curiosity, adaptability, and a drive to explore novel solutions that merge technical excellence with creativity.

EDUCATION

2023 - 2027

**Bachelor of Science (Hons)
Software Design
with Virtual Reality and Gaming**
Technological University of the
Shannon: Midlands

Expected result overall (94.83%)
Final Year Project (N/A)

2018-2023

Our Lady's Bower Secondary School

Leaving Certificate Points: **613/625**

RELEVANT WORK EXPERIENCE

12 January – 28 Aug 2026

Software Intern | Ericsson | *Cornamaddy, Athlone, Co. Westmeath*

- Started in the Orchestration Department, gaining experience in enterprise software engineering workflows.
- Later transitioned to the Cloud RAN team, working on Cloud Network rApps and supporting the integration of AI into development workflows to improve productivity and automation.
- Contributed to service migration projects, modernising applications by migrating services to improved frameworks, enhancing performance, and implementing comprehensive testing in CI/CD pipelines.

03 June – 29 Aug 2025

Summer Intern | Ericsson | *Cornamaddy, Athlone, Co. Westmeath*

- Work in the Orchestration Department, learn team management and software engineering workflow, agile methodologies.
- Was assigned tasks like migrating and improving performance of services, upgrading services and testing with Spring Boot Framework network products.

January – August 2024

Coder and Animator | Infinite Kidz Start Up

- Volunteered to develop engaging user interfaces for an educational children's platform.
- Assisted in coding features and troubleshooting issues to enhance gameplay mechanics.
- Collaborated with a small team to create prototypes and refine project deliverables.
- Provided creative input on visual design and user experience improvements.

OTHER WORK EXPERIENCE

June 2024 – now

E-Commerce Product Coordinator | Burgess Department Store | 1-7 Church Street, Athlone

- Managed website content updates, ensuring accurate product listings and promotions.
- Optimized images and layouts to enhance the site's visual appeal and functionality.
- Monitored website performance and suggested improvements for user experience.

June 2023 – June 2024

Sales Assistant | Burgess Department Store | 1-7 Church Street, Athlone

- Provided excellent customer service, assisting shoppers with inquiries and purchases.
- Handled transactions, returns, and inventory management to maintain store efficiency.
- Assisted in visual merchandising and product placement to maximize sales.
- Worked effectively as part of a team to achieve sales targets and improve customer satisfaction.

Relevant Projects (<https://github.com/JennyTran17>)

Microservices and Web Services

Offshore Wind Guardian

AI-powered monitoring system for offshore wind farms with BME280 environmental sensors, 6-axis IMU, and hydrophone acoustic monitoring. This project was researched and implemented by a small team of five members for the CleanerGrid competition.

- **Backend:** Python FastAPI with InfluxDB and Prometheus
- **Monitoring:** Grafana dashboards with Prometheus
- **Sensors:**
 - BME280 (temperature, humidity, pressure)
 - 6-axis IMU (accelerometer, gyroscope)
 - Hydrophone (underwater acoustic monitoring via USB sound card)
- **Containerization:** Podman/Docker

Grassland Resilience

A complete web application that monitors grassland health across Ireland using NASA satellite data:

- **Frontend:** Interactive map with layer controls, field management, and user authentication
- **Backend:** NASA satellite data API with real-time processing
- **Data Sources:** NDVI Anomaly, Soil Moisture, Active Fires, Risk Assessment

AI and Machine Learning

AI Security Firewall

Traditional security tools monitor network ports and user endpoints, but they have zero visibility into what autonomous AI agents are doing internally. An AI agent can browse your environment, write code, and call system APIs, and there is no way to tell if it's executing a legitimate task or has been hijacked via prompt injection.

This project solves that gap. It's a runtime security firewall for AI agents that:

1. Intercepts every system call (file I/O, network, subprocess) an AI agent makes before it executes
2. Validates each call against the user's original prompt intent, blocking anything unauthorized
3. Visualizes the entire decision flow in a real-time 3D frontend
4. Tests itself with a built-in Red Team Agent that automatically attacks the firewall to prove it works

Modelling, Simulation and Game Development (more in github)

Meteor Impact Simulator (Hackathon 25)

The application is primarily being developed in the Unity game engine, allowing us to craft intuitive, engaging 3D environments. The team is utilizing Python/other backend tools to handle precise physics and orbital mechanics calculations, ensuring the experience is both fun and scientifically sound. Our mission is to demystify meteorology and space science, making it easy, memorable, and fun to understand.

RoboQuest (Game Fleadh25)

Project made for the Game Fleadh 2025 - A short game where a broken robot is trapped on an abandoned planet in isolation, and its goal is to find ways to escape by repairing its spaceship, fleeing them off the planet somewhere into the void.

SKILLS PROFILE

Technical Skills:

IT & Programming:

Proficient in Java, Spring Boot Framework, Python, HTML, CSS, JavaScript, C#, C++, with expertise in game development (Unity) and VR/AR technologies. Internship experience emphasizes the experience in Docker/Kubernetes, CI/CD, AWS, Cloud Infrastructure, Golang.

Soft Skills:

Communication & Teamwork:

Developed strong communication and leadership skills as Vice President of the Asian Society, organizing events and coordinating diverse teams. Collaborated in hackathons (NASA Space Apps, Game Fleadh, Pro-a-thon) under pressure

to develop prototypes and pitch solutions. Gained retail and marketing experience at Burgess Department Store, assisting customers and ensuring smooth operations.

Problem-Solving & Innovation:

Designed an app prototype for students with disabilities in Pro-a-thon and contributed to a 48-hour coding sprint in NASA Space Apps Challenge. Applied analytical and troubleshooting skills in game development and volunteer coding projects at Infinite Kidz Start-Up.

Leadership & Creativity:

Managed budgets, events, and community engagement as Vice President of the Asian Society. Part of the RUNEU Student Council to help raise the voice of students and promote RUNEU principles. Developed interactive animations and assets for an educational platform at Infinite Kidz Start-Up. Featured in Study in Athlone promotional videos and recognized for creativity and technical skills with a Game Fleadh award.

Multilingual:

Fluent: English, Vietnamese | Conversational: German, Korean.

ACHIEVEMENTS

2023- Best Design Communication Graphics Project in Our Lady's Bower Secondary School

2024

Feb: Received Ericsson Summer Internship for 2025 for Leaving Cert Results from Ericsson Bursary

May: Received the National Student Award for highest mark in Design Communication and Graphics Leaving Cert Subject in All Ireland by Technoteachers.

June: Received certification for a Short-Advanced Program RUNEU about Drawing and Technology using AR in art

Oct: Participated in teams in Nasa Space Apps Challenge Hackathon for a 48 Hour coding session and web design about Exoplanets.

2025

Jan: Organized a cultural celebration for Chinese New Year and increased engagement for TUS Asian Society

Feb, Nov 2025: Received Deans Honours Awards from TUS Midlands

5 March: Won the first Game Fleadh 2025 award for Best in Game Play as Team Evolution with the game RoboQuest (currently on itch.io)

September: Achieved Certificate for RUNEU Lead Speak Success program in Finland for leadership skills.

October: Received Recognition Certificate for Nasa Space Apps Hackathon for the project Meteor Impact Simulator

2026

May: Achieved Certificate for RUNEU Video Games and Social Transformation program in Spain.

REFEREE 1

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REFEREE 2

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